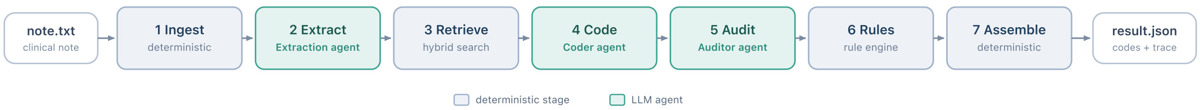
medcoder: Auditable Medical-Coding Pipeline

1 · Architecture & Data Flow

medcoder converts a free-text clinical note into billable codes (ICD-10-CM, the US clinical modification of ICD-10, for diagnoses; CPT for procedures) via a seven-stage pipeline (from `note.txt` to `result.json`) as shown in the diagram below. Three stages are LLM agents; four are deterministic code. Every stage is idempotent and writes to a trace file, so a reviewer can see exactly how each code was reached.



#	Stage	Runs on	What it does	Output
1	Ingest	deterministic	normalises text, splits SOAP sections, windows long notes, keeps offsets	clean note
2	Extract	Extraction agent	pulls facts (span, normalised term, present/absent/possible, dx/px); a NegEx backstop fixes polarity	facts
3	Retrieve	deterministic	hybrid search returns a real-code shortlist per fact (§2)	candidates
4	Code	Coder agent	picks 1 to N codes per fact, only from the shortlist, quoting evidence	assigned codes
5	Audit	Auditor agent	a second, independent model re-checks each (evidence, code) pair	agree / disagree
6	Rules	deterministic	ICD-10-CM checks (Excludes1 conflicts, missing detail, format)	typed warnings
7	Assemble	deterministic	blends signals into a high/medium/low tier; writes the result and trace	result.json

Each run writes one result file (`result.json`, or `result.md` / `result.annotated.md` per `--format`) plus a `trace.json` recording every stage's output: the facts, the per-fact shortlist, the coder choices, and the auditor verdicts.

2 · Code retrieval / Filtering strategy

Design: the coder is constrained to select only from a retrieved candidate set and can never free-generate a code. For each extracted fact, a short list of real catalog codes is retrieved; the coder may choose only from that list, and any code not on it is discarded during validation. Invalid codes are therefore impossible by construction, which matters because models that free-generate ICD-10 codes frequently emit invalid or non-billable ones [1].

The shortlist is built from two complementary searches over the code descriptions, fused into one ranked list:

- **Dense search** matches by meaning (semantic similarity via cosine), so "DM type 2" finds "type 2 diabetes mellitus". Sentence-transformer embeddings (`all-MiniLM-L6-v2` by default) are queried over a FAISS index (top 50).
- **Lexical search (BM25)** matches exact wording, where a phrase like "essential hypertension" must be read word for word (top 50).
- **Reciprocal Rank Fusion** [2] merges the two by rank, so the different scoring scales need no calibration; the top 15 become the shortlist. Each fact is also queried with a few agent-supplied synonyms (for example "MI" to "myocardial infarction") to widen recall.

Data: the real CDC ICD-10-CM FY2027 catalog (about 74,879 codes, public domain) is bundled, so the large-vocabulary challenge is genuine. CPT is AMA-copyrighted, so a clearly marked synthetic CPT-shaped catalog is shipped, and a licensed real-CPT catalog can be substituted through configuration without code changes.

3 · LLM usage & Prompting approach

Three role-specialised agents mirror the real coder-then-QA workflow, all behind one LiteLLM gateway so providers and per-agent models swap by environment variable.

Agent	Human analog	Model (default)	Key constraint
Extraction	clinical scribe	`openai/gpt-5.4-mini`	reason first, then emit schema-validated JSON

Agent	Human analog	Model (default)	Key constraint
Coder	medical coder	`openai/gpt-5.4-mini`	may only choose codes from the shortlist
Auditor	QA reviewer	`anthropic/claude-haiku-4-5-20251001`	a different model family by default

- A **different model family for the auditor** reduces correlated errors and self-preference bias, where a model rubber-stamps its own answer [3].
- **Reason first, then format:** forcing JSON-only output degrades reasoning [4], so the model thinks internally then emits one schema-validated JSON object, with a single repair-retry on a validation failure.
- **Cost discipline:** the extraction call also returns the encounter type and the retrieval synonyms in one shot; verification is selective (procedures and low-confidence diagnoses only) and calls are batched.
- **Confidence is not the raw LLM number** (which is overconfident): we blend the fused retrieval rank, the coder's discounted confidence, and an auditor adjustment, then bin into high, medium, or low tiers.

4 · Key decisions & trade-offs

Decision	Why	Trade-off
Retrieve, then constrain (shortlist, not free generation)	Removes invalid codes; turns a 75k-way generation problem into a 15-way choice [5]	Bounded by retriever recall, now measured as a separate eval stage
Coder plus independent auditor	Best precision and recall; heterogeneous checkers cut correlated errors [3]	Extra LLM calls, reduced by selective and batched verification
Hybrid retrieval with RRF	Catches exact terms and paraphrases, no score tuning	Two indexes to build (about a minute on 75k codes)
Real ICD-10-CM plus synthetic CPT	Meets the large-vocabulary challenge within AMA licensing	CPT demo runs on synthetic codes
Deterministic rules and temp-0 pinned models	Auditable, reproducible, no LLM cost for hard checks	Curated rule subset; foregoes a small sampling-accuracy gain

5 · Limitations & extensions

- *Assistive, not autonomous.* On the evaluation set the system reaches ICD-10 micro-F1 about 0.5; full-vocabulary ICD-10 coding is hard in general (reported state of the art is around 0.54 [6]), so a human reviewer is required. The output (evidence spans, confidence tiers, editable fields) makes that review fast.
- *CPT is synthetic.* Real-CPT accuracy must be re-validated on a licensed catalog, which is substituted through configuration without code changes.
- *General-purpose embedder.* `all-MiniLM-L6-v2` is the default because it is local, keyless, fast, and adequate for the short code-description strings being embedded. Production clinical text would benefit from a domain-specific biomedical embedder (SapBERT or PubMedBERT, trained for mention-to-concept linking); the embedder is pluggable via configuration.
- *Evaluation is directional.* The small authored gold set (n=4) gives directional signal only; the contribution is the metric methodology (micro P/R/F1, hierarchical F1, per-stage retrieval recall), not the absolute scores. A larger synthetic set would not add credibility; real validation needs a licensed labelled corpus such as MIMIC-IV. Precision is over-coding-bound (more codes emitted than gold labels), so the main lever is a more selective coder.
- *Extensions the architecture supports but does not yet implement.* A Postgres-backed retriever (semantic, full-text, and fuzzy search in one datastore), biomedical embeddings with a SNOMED-to-ICD crosswalk, a fuller rule engine (ICD-10 tabular plus CMS NCCI), and a FastAPI reviewer UI. Idempotent stages can be orchestrated by Airflow or Celery.

References

[1] NEJM AI, 2024 (LLMs free-generate invalid ICD-10 codes). [2] Cormack et al., 2009 (Reciprocal Rank Fusion). [3] arXiv:2410.21819 (heterogeneous verifiers cut correlated errors). [4] arXiv:2408.02442 (JSON-only output constraints degrade reasoning). [5] arXiv:2407.12849 (retrieve-then-rerank constrained coding). [6] RAG-Coding, 2026 (full-vocabulary ICD-10 state of the art about micro-F1 0.54).